Block Diagram

Team 2_muzakker_3

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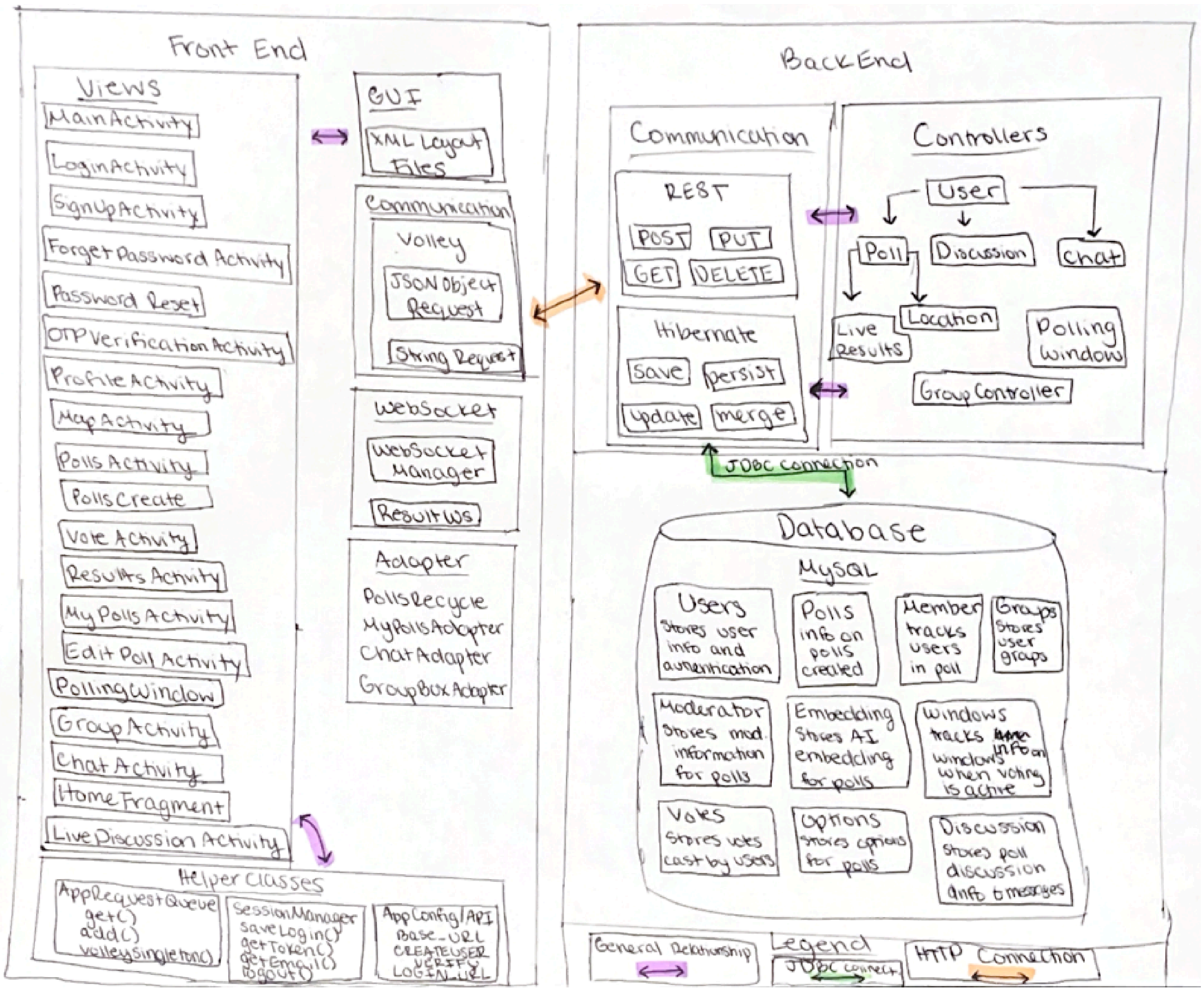
Smart Polling and Voting App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Polls Create
Vote Activity
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stores user groups
Moderator
stores mod. information for polls
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stores AI embedding for polls
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stores options for polls
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stores poll discussion info + messages
Legend
General Relationship
JDBC connect
HTTP Connection

Frontend (Currently Implemented):

I implemented three main frontend features in the Android app: Map/Geofencing, Signup + OTP verification flow, and Chatbot UI.

Map/Geofencing: The MapActivity shows a Google Map with the ISU campus geofence drawn as a circle. It requests location permission, gets the user's current GPS location, and displays the user marker on the map. The activity calculates distance from campus center locally and also sends latitude/longitude to the backend (/location/eligibility) using Volley to confirm eligibility. Based on these checks, the "Continue" button is enabled only when the user is eligible to vote.

Signup + OTP verification: The SignupActivity handles new user registration. It collects name, email, and password, validates input, and sends a POST request to the backend (/users) using Volley/JSON. If signup succeeds, it automatically navigates the user to OtpVerificationActivity, where the user enters the email verification token to activate the account.

Chatbot: The ChatActivity provides the chatbot screen using RecyclerView and a chat adapter. When the user sends a message, it immediately shows the user bubble, then calls the backend chatbot API using Retrofit. If the user is logged in, it uses the authenticated endpoint with JWT; otherwise it falls back to the public chatbot endpoint. The response from the server is displayed as the bot message in the UI.

Backend:

Controller Documentation

UserController: manages all user accounts including registration, login, logout, email verification, token handling, password reset, and basic CRUD operations

PollController: handles creating polls of 4 different types, casting and deleting votes, validating user location, retrieving poll statistics, managing moderators, joining private polls, controlling poll visibility, and broadcasting results

LocationController: verifies if a users coordinates fall inside the campus geofence so that only users that are on the ISU campus can vote

LiveResultsController: handles live poll updates through WebSockets by sending initial poll results when a user subscribes and broadcasts updated results whenever a vote gets added, changed, or deleted

DiscussionController: manages creating discussions, joining/leaving them, viewing details, sending and receiving messages, and closing discussions so that users can access an interactive conversation system

ChatController: handles AI chat bot features, lets users fetch chat responses, and retrieve personal chat history and message statistics

GroupController: manages user created groups that can be used inside Polling Windows to restrict voting access

PollingWindowController: manages time based voting restrictions where poll owners can create, open, and close windows

Table Relationships Diagram